

Design of Electronic Toll Collection System based on Global Positioning System Technique

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Abstract—Currently, most electronic toll collection (ETC) systems around the world are implemented by DSRC (dedicated short range communication) technology. However global positioning system technique is applied to ETC to replace DSRC technique in recent years. It is an innovative technology for Expressway Network electronic toll collection solution, which is totally different scheme comparing to traditional DSRC-based technology. In this paper, the frame composing and working flow of the system are described, the design of GPS-based ETC system are discussed detailedly, and the advantages and disadvantages of the system are discussed at last.

Keywords—*Electronic Toll Collection; Global Position System; system frame; module design*

I. INTRODUCTION

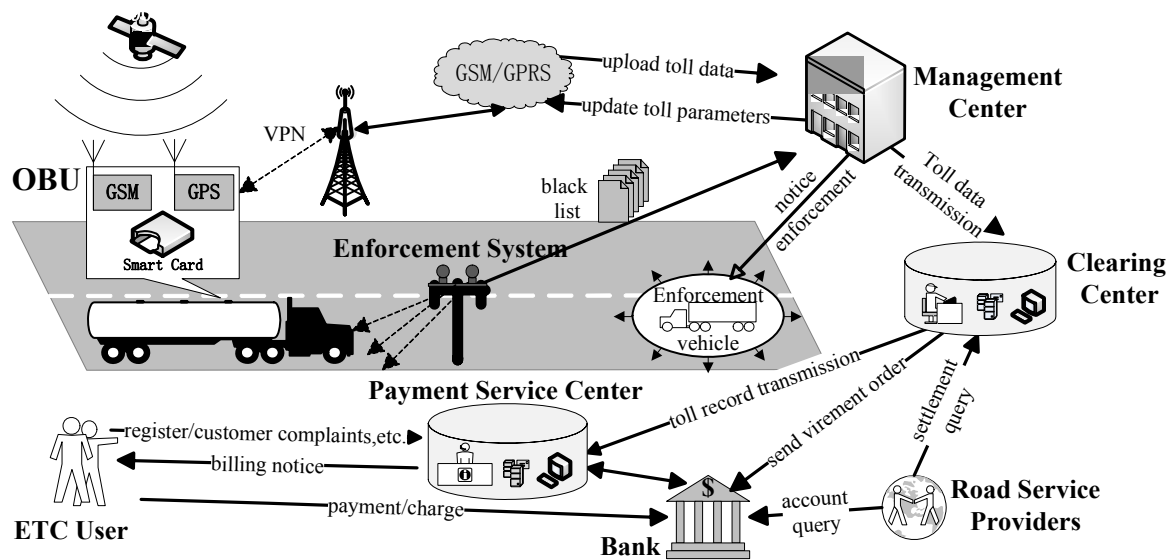
As one of the major research topics in intelligent transportation system (ITS) electronic toll collection system is considered as an effective method in order to alleviate traffic congestion and jams, enhance the convenience and safety of travelers, and minimize air pollution and fuel consumption for environmental protection need. ETC system determines whether the vehicles passing are enrolled in the program, alerts enforcers for those that are not, and debits electronically the accounts or the amount in the IC card of registered cars without their stopping [1]. At present the traditional technologies used in ETC system are focused on as DSRC (dedicated short range communication) system since the on board unit (OBU) installed in the vehicle can only communicate with the road side unit (RSU) within a short range area. However a new generation of ETC is rapidly developed to replace DSRC-based ETC systems.

Global positioning system (GPS) technology has become the new trend for road charging system, which implements ETC system based on positioning and Global System for Mobile communication (GSM) technologies. The OBU receives vehicle positioning information from GPS, automatic calculates and deducts the road using charge, and sends fee data to the Control Center through the GSM.

II. SYSTEM ARCHITECTURE

The system architecture of GPS-based ETC system is illustrated in Fig. 1, which includes five key components: OBU, Enforcement System, Management Center, Clearing Center and the Payment Service Center.

The working process of GPS-based ETC system are described as follows procedures: (1) Drivers goes to the payment service center registers and installs OBU and applies for pre-paid card or post-paid card. (2) When vehicle moves into the charging zone, OBU compares current vehicle position coordinate from GPS with the virtual toll node coordinate kept in the storage of OBU. After logistic determining, the OBU sets up wireless communication channel through GSM module. (3) OBU sends transaction message to management center system by the GSM module through mobile network. (4) After auditing the management center saves toll data and sends back transaction information to the OBU. (5) The OBU receives and displays the transaction result. If error transaction result is replied the OBU is abnormal payment state, the OBU is still working. (6) When the vehicle moves out the charging zone, if the OBU has abnormal state or the vehicle doesn't install an OBU the violation will be processed.



(7) The clearing center clears all toll data from management center, and divides in road service providers. (8) The Payment Service Center collects toll records and clearing data for account query.

III. SYSTEM COMPONENTS DESIGN

A. OBU

The OBU includes several components: Integrated GPS and Dead Reckoning (DR) positioning module, GSM module, Micro-programmed Control Unit (MCU), Smartcard module and human interface, as shown in Fig. 2.

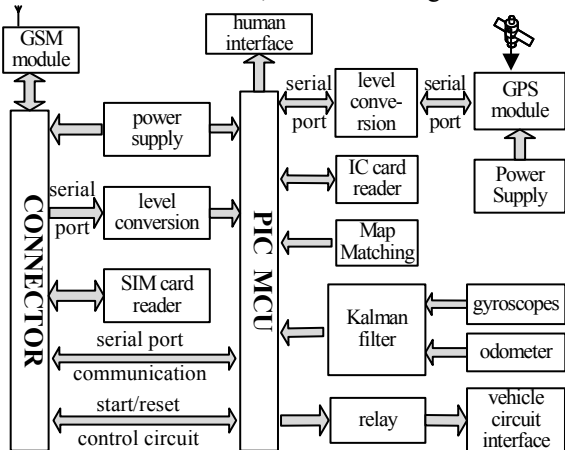


Figure 2. Module component of OBU.

1) *Integrated GPS&Dead Reckoning (DR) Positioning Module:* In order to further improve the positioning accuracy, DR was be used for GPS positioning error compensation.

2) *GSM Module*: The OBU and management center system communicate each other through the dedicated VPN (virtual private network) based on GSM/GPRS mobile communication network.

3) *Micro-programmed Control Unit*: runs the GPS client program and controlling all modules.

4) *Smartcard Module*: The vehicle drivers can pay for in advance of the journey (pre-paid card) or after the journey (post-paid card). The smartcard also plays an integral role in the security features of the OBU.

5) *Human Interface* : Indicates the OBU status, vehicle information, card residual amount etc.

The working principle of the OBU is described as follows: The highway network is divided into a number of independent sections by entrances, exits and alternative interchange. Some points in sections are defined as virtual toll nodes and its coordinate are stored in the OBU memory. When vehicle moves into the charging zone the vehicle's coordinate from Integrated GPS and DR positioning module are compared with the virtual toll node coordinate. If it is matching then marks this node as "1#" else gives up. Repeat comparing until it is matched again and mark the node as "2#". Through the two nodes we mark this section as "SECTION A". According to the distance and the fee rate of "SECTION A" the OBU computes the payable toll. By GSM module the OBU sends toll data to the Management Center, receives transaction information and display in the screen of OBU.

B. Enforcement System

The enforcement system provides a mechanism to deal with the violation. The design objective is that it should catch the license plate pictures of vehicles and recognize the license number in nearly real time. Enforcement system includes fixed enforcement and moving enforcement.

1) *The Fixed Enforcement:* There are monitoring stations built in the expressway. When monitoring station detects the vehicle entering the enforcement line the camera catches the vehicle license plate correctly and automatic recognizes the license plate number by OCR (Optical Character Recognition) in real time. When the vehicle gets closer to

monitoring station the DSRC module read the vehicle OBU status and registered information. If the license number recognized is matched to the registered and the status of the OBU is normal then deletes the images, else sends the images to the management center database for violation processing. If monitoring station doesn't detect the vehicle OBU then sends the images to the tollgate for manual toll collection.

2) *The Moving Enforcement*: Inspectors use the DSRC in the monitoring vehicle to read the OBU status and registered information of passing vehicle. By querying the registered user database in the in-vehicle computer if the vehicle actual information isn't match to the registered information then order the vehicle to park for violation processing.

C. Management Center

The management center is the core part of the system and its functions includes as follows:

- Manage and change the geographical scope of charging zone such as adds or deletes a toll road.
- Set up toll road fee rate and digital map data, etc.
- Collect and save toll data for clearing.
- Have extended function such as vehicle navigation, fleet management, emergency assistance, road information service, traffic flow analysis, etc.

The architecture of management center is shown in Fig4. Communication server takes charge of connect wireless communication network and internet so that monitoring server can exchanges data with the OBU, enforcement system and clearing center. On the one hand, OBU sends transaction message to mobile communications short message center and communication server read message data through communication interface. On the other hand, by communication protocol conversion the monitoring server sends command from client to GSM network. OBU and moving monitoring vehicle receive message data and perform the corresponding action. Toll data in database server is uploaded to clearing center by TCP/IP of the communication server.

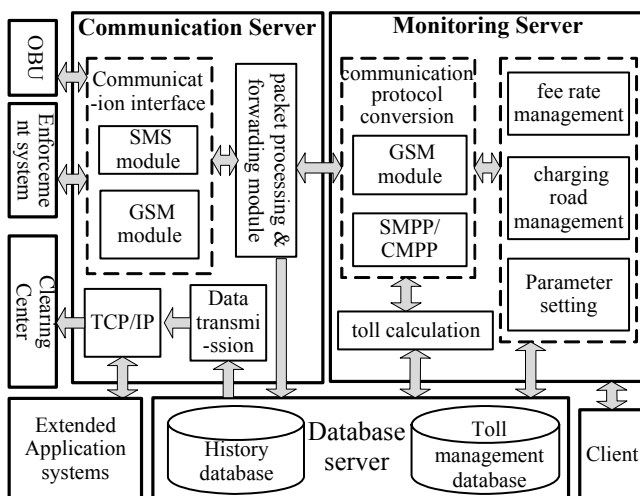


Figure 3. Module component of Management Center.

D. Clearing Center

There are several design principles to be considered:

- Unified toll data collection.
- Unified clearing account management.
- Unified settlement time.
- Unified clearing money virement.

The clearing center clears toll data, calculates the money and sends virement command to bank. According to the command received, bank transfers money from the clearing center account to each road service provider's account. The subsystems of the clearing center and its functions are summarized as follows:

- 1) *Data Collection Subsystem*: Collect toll data collection, check the validity, continuity and integrity.
- 2) *Automatic Clearing Subsystem*: Deal with toll data, transfer money to each road service provider.
- 3) *Daily Settlement Subsystem*: Transfer amount of money from clearing center to each road service provider.
- 4) *Accounting Management Subsystem*: Deal with account of expressway network toll collection.
- 5) *Reports Subsystem*: Make various reports.
- 6) *Bank Interface Subsystem*: Transmit command data between clearing center and bank.
- 7) *Data Backup and Recovery Subsystem*: Backup and recovery data of clearing system.
- 8) *System Management Subsystem*: Setting system parameters, toll parameters and time synchronization.
- 9) *Operation Statistic and Query Subsystem*: Data count and information query.

E. Payment Service Center

The payment service center takes charge ETC users registration, smartcard issuing and customer service. The subsystems of the payment service center and its functions are summarized as follows:

- 1) *Encryption Subsystem*: Manage, issue and delete the encryption of ETC.
- 2) *Issuing Subsystem*: Issuing smartcard and OBU.
- 3) *Customer Information Management Subsystem*: User registration, account recharge, consumption information query, bill printing service and arrearage notification.
- 4) *Bank Interface Subsystem*: Transmit command data between payment service center and bank.
- 5) *Clearing Center Interface Subsystem*: Collect toll data from clearing center.
- 6) *Customer Call Subsystem*: Hot line consultation and complains management, short message service.
- 7) *Web Site*: Registered users modify personal profile or list personal transaction history by logging the web site.

IV. CHARACTERISTICS OF GPS-BASED ETC

GPS is an evolutionary technology for ETC as compared to DSRC and it has advantages that DSRC doesn't have. However, there are still some problems of GPS solution.

A. Advantages

- The total capital input of road side unit will be largely reduced.
- It's easy to setup a new toll area or remove the old ones.
- By calculating the traveling distance the vehicle traveling multi-path identification problem will be completely resolved. In the DSRC-based ETC, that will be a problem.
- The interconnection problem of different DSRC provider can be broken by VPS solution because VPS adopts the worldwide standard.
- It has expansibility once other ITS application systems are integrated into GPS-based ETC.

B. Disadvantages

- GPS is a new coming technology; it is currently on the development and field trial stage worldwide. The business operation instance is only in Germany.
- It is more difficult in the matching process between the debit and enforcement information, so the system needs more post-processing jobs in order to reduce the mismatch failure.
- The cost of OBU is still high for the users.

DSRC and GPS technology has its own characteristics. For a relatively independent of toll road such as bridge or tunnel, DSRC-based ETC system is probably the best choice. However for a larger toll road network, GPS-based ETC system is more suitable.

V. CONCLUSION

With the development and improvement of GPS-based ETC system, along with its flexible definiens of charging zone and charging mode, better compatibility with other system in ITS and other advantages that DSRC-based ETC system lacking. We believe that in the near future, GPS-based of ETC will be widely used in internal road charging. In this paper we described the frame composing and working flow of the system, discussed the design of GPS-based ETC system components detailed. The application of GPS-based ETC in internal expressway network toll collection will be the major issues of our future work.

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